TRANSISTORLAB

FROM SILICON TO INTELLIGENCE

Final Project Report

Project	TransistorLab: From Silicon to Intelligence
Project type	Interactive web-based educational simulator
Team size	5 members
Prepared by	[Enter team member names and roll numbers]

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Abstract

TransistorLab is a web-based educational project designed to demonstrate how transistor-level switching connects to digital logic, arithmetic circuits, analog amplification and simplified neural models. The project is organised into three modules. The Digital module demonstrates a BJT switching model, NOT/NAND/NOR logic operations and a 4-bit binary adder. The Analog module demonstrates a common-emitter amplifier with gain and output clipping. The Neural module demonstrates an artificial neuron using step or sigmoid activation and a leaky integrate-and-fire (LIF) neuron using a simple time-stepped model. The models are implemented in JavaScript and presented through an interactive single-page website. The report describes the design approach, governing equations, testing plan, limitations and team responsibilities. Actual measured and simulated results should be entered in the validation tables after testing.

1. Introduction

A transistor is a fundamental electronic device that can operate as a switch or as an amplifier. When used as a switch, transistors form the basis of digital logic circuits. Logic gates can be combined to construct arithmetic circuits such as adders. When biased in the active region, a transistor can amplify small input signals. The behaviour of electrical systems can also be represented using simplified mathematical models, including models inspired by biological neurons.

The central idea of TransistorLab is: “Transistors switch, gates compute, and adders perform arithmetic; the same transistor can amplify signals, while mathematical models can demonstrate neuron behaviour.”

2. Objectives

- Demonstrate the cutoff, active and saturation regions of a BJT using an adjustable input voltage.
- Show the operation of NOT, NAND and NOR logic gates and the ON/OFF states of their transistors.
- Demonstrate 4-bit binary addition using full-adder stages and visible carry propagation.
- Illustrate common-emitter voltage amplification and clipping at the supply limits.
- Demonstrate an artificial neuron using weighted inputs, bias and step or sigmoid activation.
- Simulate a leaky integrate-and-fire neuron and display its membrane-voltage trace and spike times.
- Compare selected model outputs with hand calculations and, where available, LTspice or breadboard results.

3. System Overview and Technology

The project is designed as a single-page web application. The user interacts with sliders, input controls and plots. The calculations are performed in JavaScript in the browser; a server-side backend is not required for the planned demonstrations. Chart.js can be used to display transfer curves and time-domain waveforms. The website may be deployed using GitHub Pages or Netlify.

Component	Purpose
HTML and CSS	Structure and presentation of the interactive page

JavaScript	Numerical models and interactive behaviour
Chart.js	Plots for transfer characteristics and waveforms
GitHub Pages / Netlify	Hosting and sharing the live website

4. Module 1 – Digital Electronics

4.1 BJT switching model

The BJT is represented using a simplified piecewise switching model. For the planned example, the supply voltage is 5 V, the base resistance is 10 kΩ, the collector resistance is 1 kΩ, the current gain β is 100, the base-emitter turn-on voltage is approximately 0.7 V and the saturation collector-emitter voltage is approximately 0.2 V.

In cutoff, the base-emitter voltage is below the assumed turn-on level and the transistor is treated as OFF.

In the active region, the approximate base and collector currents are:

$IB = (Vin - 0.7) / RB$

$IC = \beta \times IB$

The collector output is estimated from the collector resistor and is limited by the saturation condition. The output voltage is approximately $VCC - IC \times RC$, with VCE not allowed to fall below the assumed saturation value of 0.2 V. The model is an educational approximation, not a complete transistor device model.

4.2 Logic gates

The Digital module demonstrates NOT, NAND and NOR operations for binary inputs. The interface is intended to display the logic output and the ON/OFF state of the transistor or transistors used in each simplified gate representation.

Gate	Boolean expression	Output behaviour
NOT	Y = NOT A	Output is the complement of A
NAND	Y = NOT (A AND B)	Output is 0 only when A and B are both 1
NOR	Y = NOT (A OR B)	Output is 1 only when A and B are both 0

4.3 Four-bit binary adder

The 4-bit adder adds two 4-bit unsigned binary numbers. It is constructed from full-adder stages. Each stage receives one bit from each input and the carry from the previous stage. It produces a sum bit and a carry for the next stage. The final result consists of four sum bits and a possible final carry.

For a full adder:

Sum = A XOR B XOR Cin

Cout = (A AND B) OR (A AND Cin) OR (B AND Cin)

The interface should display the carry propagation through the stages so that the user can follow how the binary result is produced.

5. Module 2 – Analog Electronics

5.1 Common-emitter amplifier

The Analog module demonstrates the phase inversion and voltage amplification of a common-emitter amplifier. The user can adjust the input amplitude and gain. A two-cycle sinusoidal input is generated and the output is calculated by applying the gain and limiting the waveform to the selected supply rails.

A simplified model is:

$vin(t) = A \sin(2\pi ft)$

$vout(t) = VQ - Av \times vin(t)$

The output is clipped when it reaches the lower or upper voltage limit. The negative sign represents the phase inversion of a common-emitter stage. For a suitable circuit with an unbypassed emitter resistor, a first-order midband gain estimate may be $Av \approx -RC/RE$. The actual gain depends on the bias point, transistor parameters, load and small-signal conditions.

6. Module 3 – Neural Models

6.1 Artificial neuron

The artificial neuron calculates a weighted sum of its inputs and adds a bias. The result is passed through an activation

function. The planned interface provides three inputs, corresponding weights, a bias and a choice of step or sigmoid activation.

$z = x1w1 + x2w2 + x3w3 + b$

Step activation: output is 1 when $z \geq 0$, otherwise 0.

Sigmoid activation: $\sigma(z) = 1 / (1 + e^{(-z)})$.

6.2 Leaky integrate-and-fire neuron

The LIF model represents a neuron membrane as a capacitor with a leakage resistance. Its voltage changes over time according to the input current and leakage. The differential equation is:

$C \times dV/dt = I(t) - V/R$

The equation is evaluated numerically using Euler time steps. When the membrane voltage reaches the threshold, a spike time is recorded and the voltage is reset to 0 V. The interface displays the voltage trace and spike markers.

For a constant input current I and a reset voltage of 0 V, the idealised time to reach threshold Vth is $T = RC \ln(IR / (IR - Vth))$, provided IR is greater than Vth. This expression can be used to check the simulated inter-spike interval under constant input.

7. Testing and Validation Methodology

Validation is performed by comparing the JavaScript model outputs with independent hand calculations. Where tools and laboratory access are available, selected circuits are also compared with LTspice and breadboard measurements. All actual results must be recorded from the completed implementation; values must not be assumed or fabricated.

Test area	Validation method	Evidence to attach
BJT switching / NOT	Check cutoff, turn-on and saturation values against hand calculations; compare with LTspice DC sweep.	Transfer curve, input/output readings, LTspice screenshot
NAND and NOR	Test every possible binary input combination against the truth table.	Completed truth tables and website screenshots
4-bit adder	Run ten input pairs and compare sum bits and final carry with hand-calculated results.	Ten-case test table and adder screenshot
Common-emitter amplifier	Compare measured/simulated input and output amplitudes and gain.	Waveform screenshot and gain calculation
Artificial neuron	Check weighted-sum and activation outputs for selected input/weight/bias combinations.	Test cases and interface screenshot
LIF neuron	Compare threshold crossing and spike interval with the analytical expression for constant input.	Voltage trace, spike times and calculation

8. Validation Results – To Be Completed

Enter the actual outputs after running the final website and simulations. If a test was not performed, write “Not performed” and briefly state why.

Test	Hand calculation	JavaScript simulation	LTspice	Breadboard
NOT switching point	[Enter value]	[Enter value]	[Enter value / N.A.]	[Enter value / N.A.]
NOT saturation point	[Enter value]	[Enter value]	[Enter value / N.A.]	[Enter value / N.A.]
Adder test cases	[Attach 10 cases]	[Enter pass/fail]	N.A.	N.A.
Amplifier gain	[Enter value]	[Enter value]	[Enter value / N.A.]	[Enter value / N.A.]
LIF firing interval	[Enter value]	[Enter value]	N.A.	N.A.

9. Results and Discussion

[Complete this section after testing. Describe whether the BJT model displayed the expected operating regions, whether all logic-gate outputs and adder results matched their truth tables, whether the amplifier waveform showed phase inversion and clipping, and whether the neural models responded correctly to changes in their controls. Mention any mismatches and how they were corrected.]

10. Limitations

• The BJT switching model is piecewise and does not include all physical transistor effects, temperature dependence or detailed device parameters.

• The logic-gate transistor representations are simplified educational models and may not reproduce every detail of a practical transistor network.

• The 4-bit adder is limited to two 4-bit inputs and does not provide a general-purpose calculator.

• The amplifier uses a simplified gain-and-clipping model; it does not fully model bias networks, frequency response, loading, distortion or transistor nonlinearities.

• The artificial neuron and LIF neuron are mathematical demonstrations and do not reproduce the full complexity of biological neurons.

• Simulation results depend on the assumptions, parameter values, numerical time step and implementation used.

11. Conclusion

TransistorLab brings together three important ideas in electronics and computation: transistor switching for digital logic, transistor amplification for analog signals, and mathematical neuron models for neuron-inspired computation. The interactive demonstrations are intended to help users connect circuit behaviour with equations, truth tables and waveforms. The project also provides a practical way to compare simplified models with hand calculations and circuit simulations. Final conclusions about accuracy should be made after the validation results and screenshots have been added.

12. Team Contributions

Team role	Responsibility	Member name
Project lead / model developer	Develop six JavaScript model functions, perform hand calculations, validate outputs and approve scope.	[Enter name]
Web developer	Build and deploy the interactive single-page website; integrate models; record backup demo.	[Enter name]
Helper A – Simulation	Run LTspice tests for NOT gate and amplifier; build and measure NOT gate on breadboard if lab access is available.	[Enter name]
Helper B – Content and slides	Prepare short module explanations and presentation slides.	[Enter name]
Helper C – Testing and report	Prepare truth tables and test cases, perform hand calculations, click-test modules and assemble the report.	[Enter name]

13. References

1. Course lecture notes and laboratory instructions for digital electronics, analog electronics and neural models.
2. TransistorLab project specification and team sprint plan, dated 2 November 2026.
3. LTspice simulation files and screenshots generated by the project team (add filenames/details after testing).
4. Chart.js documentation, if used for plotting: https://www.chartjs.org/docs/latest/

Appendix A – Items to Attach Before Submission

• Screenshot of the live Digital module (BJT transfer curve, gates and 4-bit adder).

• Screenshot of the Analog module showing input and clipped output waveforms.

• Screenshot of the Neural module showing artificial-neuron output and LIF voltage trace.

• Completed ten-case 4-bit adder validation table.

• LTspice NOT-gate DC sweep screenshot and readings, if performed.

• LTspice amplifier waveform screenshot and gain readings, if performed.

• Breadboard photograph and measured readings, if performed.

• Live website URL and link or filename for the two-minute backup demo video.

Editing note: Replace all bracketed placeholders with the actual team details and measured results. Remove any validation methods that were not part of the final implementation, or mark them as not performed.